

Instructor: Date:

Learner: Regional Branch:

Program: Center:

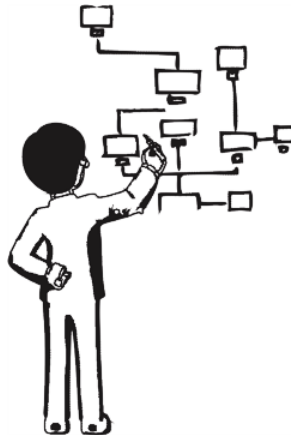
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Let's Start!

Combining Different Styles to Create the Best One!



Learning Outcome:

Build the application installation program, using IT development tools available in the market, based on solution architecture features. Recognizes and implements the different types of architecture styles that can be included in the IT solution.

Can Do Statements:

- I can talk about table driven interpreters' architecture style.
- I can explain other familiar architecture styles.
- I can describe heterogeneous architecture styles.

Ask your partner:

- What do you think it's the benefit of combining different architecture styles?
- What could be the difficulties of these kinds of combinations?





Let's Explore!

Samantha is ready for her job interview.



Instructions: Read the text below.

Martin: We're looking for someone with a broad knowledge about certain architecture styles. Do you consider yourself an expert on this matter?

Samantha: Definitely, I have been designing programs for three years and I have developed the necessary skills to work with architecture styles.

Martin: Excellent. Can you talk to me about the table driven interpreters?

Samantha: Table driven interpreters are used to build virtual machines. They include an interpretation engine and a pseudo-program which is interpreted by the engine. Such engine includes a definition of the interpretation and the current state of the process. This style has four components which are the interpretation engine, a memory, a representation of the interpretation, and another representation of the current state of the program.

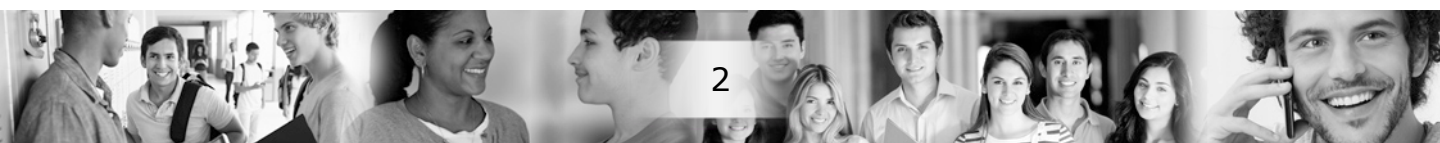
Martin: Good, now can you explain to me other familiar architecture styles?

Samantha: Well, there are different types, some are commonly used and some others are very particular. For instance, distributed processes. They have been facilitating inter-process protocols and they have helped more communication software to appear. Another example is main programs/subroutine organizations that mirror the language in which the system was written and the subroutines provide control. We also have domain-specific software architectures to obtain references and state transition and process control systems to move among systems and control them.

Martin: Terrific! Finally, describe to me the heterogeneous architecture style.

Samantha: This style implements more than one set of instructions where there's only one memory. A mixture of architectural connectors is used to combine different styles to improve power, efficiency, performance, programmability, and portability. The idea is to organize the components by hierarchy. This style has been growing to simplify processes as it has contributed to most application designers to implement their software.

Martin: You're hired!





Let's Practice!

This is a little explanation about heterogeneous architecture styles.

Instructions: Use the words from the word bank to complete each sentence. Practice reading the sentences aloud. Check your pronunciation with your instructor.

1

- Is explained
- Is explain
- Are explaining

2

- Are implementing
- Is implemented
- Are implement

3

- Are using
- Is used
- Is use

4

- Must be organize
- Must be organized
- Must be organizes

5

- Are used
- Is use
- Is used

1. The concept of heterogeneous architecture styles ____ ____ by Samantha to Martin.
2. According to Samantha, more than one set of instructions ____ ____ in this kind of style.
3. A combination of architectural connectors ____ ____ to associate different styles to improve power in the system.
4. The components in this style ____ ____ in a hierarchical manner.
5. This style ____ ____ by many people because it has contributed to most application designers.



Let's Practice!

I have been analyzing some concepts and I have understood other familiar architecture styles.

Instructions: Proofread the sentences. Correct the mistakes.

1. I has been reading about other familiar architecture styles.

2. I have been read about distributed process.

3. I had realized that this is another architecture style.

4. I have been search for information about distributed process.

5. I have being studying about domain-specific software architecture.



Extension Activity

There are a lot of styles!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about other familiar architecture styles.
- Take notes about those styles.
- Write a description for each familiar architecture style.
- Make a chart to compare the different architecture styles you have found.
- Show your chart to your classmates and teacher.

Self-Assessment

Yes

No

- I search information about other familiar architecture styles.
- I summarize the information I found.
- I write a description for each familiar architecture style.
- I make a chart to compare the other familiar architecture styles.
- I share the information with my class.

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Target Words

- **heterogeneous** (adjective): made of different parts
e.g. Describe to me the heterogeneous architecture style.
- **pseudo** (adjective): almost similar to the referent
e.g. They include an interpretation engine and a pseudo-program which is interpreted by the engine.
- **set** (noun): a group of items
e.g. This style implements more than one set of instructions where there's only one memory.
- **broad** (adjective): that is wide
e.g. We're looking for someone with a broad knowledge about certain architecture styles.
- **simplify** (verb): to do things easier to do
e.g. This style has been growing to simplify processes.
- **mixture** (noun): a combination of things unified
e.g. A mixture of architectural connectors is used to combine different styles.



Learning Points

Present Perfect - Affirmative Form

You use the **present perfect tense** to talk about actions that took place in the past and they still happen in the present.

Whenever we use **present perfect tense** in affirmative sentences, we are usually talking about experiences or relevant actions from the past that continue in the present. We use **have** (**has** the for 3rd person) as an auxiliary and the following verb goes in **past participle**.

Let's review the grammatical structure:

Subject + **have/has** + **verb in past participle** + complement

Examples:

*Heterogeneous styles **have changed** the ways of structuring processes in a system.*

*The table driven style **has included** a system for self-interpreting.*

Examples

- *I have developed the necessary skills to work with architecture styles.*
- *They have helped more communication software to appear.*
- *It has contributed to most application designers to implement their software.*

Table Driven and Heterogeneous Architecture Styles

The **table driven architecture style** is known for applying virtual machines for interpreting, and the **heterogeneous architecture style** is able to combine a series of methods for a more complete approach.

Table driven architecture: By using interpreters virtual machines are built to connect the semantics of the program to the engine.

Heterogeneous architecture: Instead of being an individual pure method to structure processes, this style of architecture offers the combination of different styles by using hierarchy.

In the next chart we will be able to see the main concepts of these two styles.

Examples

- *Table driven interpreters are used to build virtual machines. They include an interpretation engine and a pseudo-program which is interpreted by the engine.*
- *A mixture of architectural connectors is used to combine different styles to improve power, efficiency, performance, programmability, and portability.*

